

Impact of Gender-Difference of Emotional Attachment on Relationships

Prutha B. Patel

Department of Psychology, UC Davis

PSC 103A - Statistical Analysis of Psychological Data

Dr. S. Blozis Villarreal

March 14, 2025

Impact of Gender-Difference of Emotional Attachment on Relationships

One of the topics in the field of psychology and sociology aims to understand how humans interact with each other to form relationships and how one maintains these relationships. To understand this topic we would need to examine the fundamental aspect of these relationships. One of the foundational psychological characteristics of a relationship is emotional attachment. Emotional attachment can help us understand romantic relationships, family dynamics and societal functioning. Along with understanding the psychological characteristics of emotional attachment for relationships it is further interesting to examine and understand the emotional attachment between females and males. Recent studies indicate that women show stronger emotional expressions than men. Further studies also show that happiness, sadness, and fear are more typically associated with women, and on the other hand anger and pride are more typically associated with men (Fischer & LaFrance, 2015). Although the characteristics of emotional attachment of a person isn't a defined numerical unit we can perform surveys, self-reports questionnaires, and other statistical methods to help us understand the gender difference of emotional attachment.

To understand the gender difference of emotional attachment we can look at studies that explore the relationship between parents and their infants. In a randomized crossover study, 62 preterm neonates were held in a kangaroo care or also commonly known as skin-to-skin contact (Johnston, 2011). The infants were held for 30 minutes before and during the heel lance procedure with either the mother or the father of the infant. In this study it was observed infants in the maternal kangaroo care had displayed statistically significantly lower scores on pain response of infants at 30 and 60 seconds after the heel lance procedure compared to when the infants were in the paternal kangaroo care. Mothers of the infant were more effective in

decreasing the pain response than the father of the infants. Another study surveyed 250 college students from the University of Porto, Portugal with 61 females and 89 males through a subjective form of assessment on love. It was observed that women displayed a statistically significantly higher estimate for passionate love, compassionate love, intimacy and commitment as compared to men with a $p\text{-value} < 0.05$. This study examined how females and males estimated self and other types of love (Neto, 2023). In another study, 2570 college students from seven different countries Egypt, Algeria, Kuwait, Lebanon, Iran, India and Turkey were surveyed (Abdel-Khalek et al., 2022). The college students were given a 16 survey questionnaire on Love of Life. The study aimed to compare any difference among the seven countries and examine differences for Love of Life in terms of sex-related differences. The results of the study provided evidence for sex difference in the scores for love of life. There was a higher mean score for women than the male in each country with a $p\text{-value} < 0.001$. Further, the difference between the total mean score for the love of life between all men and all women showed a statistically significant higher for women. These studies further build on our hypothesis supporting evidence indicating a difference in emotional attachment between females and males. This can help us understand how relationships are built between people within the same gender and between genders.

With the support of these studies as evidence I am further interested to understand the pattern in my data set with 170 participants from MIDUS 3 surveys. To understand my data set in relation to my research question I would perform a two-independent sample t-test to test my hypothesis if women do have a significantly higher emotional attachment than men. Performing this t-test would give us a better understanding of whether our dataset follows the trend of

women having a higher emotional attachment or the emotional attachment is not significantly different between the women and men.

Method

Participants

The data set that we will be performing our hypothesis on is from a subset date of MIDUS 3. MIDUS is a national survey collecting data of Americans aged between 25-75 years. In this national survey respondents were asked to provide detailed information on their physical and mental health. MIDUS 3 is a third wave of survey data collected in the year 2013. This study includes longitudinal follow-up from MIDUS which included 7,108 participants from a random digit dialing sample, oversamples from five metropolitan areas, siblings from the random digit dialing samples and a national random digit gosling sample of twin pairs. Total of 3,294 out of 4,460 cases had completed the for MIDUS 3 data collection. For our dataset we will be working with 170 participants that have completed the survey. For our data set we will be using the subset of the MIDUS 3 of 170 with 97 female and 73 male participants for our hypothesis testing gender difference of emotional attachment.

Procedure

Participants in the MIDUS 3 were first telephone interviewed for 45-minutes, and then invited to complete a 100 page mail questionnaire and later followed up for a 25-minute cognitive interview. From May, 2013, through November, 2014 2,732 participants had completed and returned their self-administered questionnaires (SAQs). The questionnaire included writing their date of birth to verify the date of birth given by the participants during the telephone interview. MIDUS first sent out an advance letter explaining the three procedural rounds of data collection with a \$2 pre-incentive. Another \$25 incentive was sent after the participants

responded back within one week of completing the telephone interview. The questionnaire packet containing the questionnaire booklet along with two five dollar bills as pre-incentive were sent out to the participants once they completed the telephone interview. In the SAQ of emotional attachment clear instructions were written on the booklet for the participants as to how to fill out the questionnaire. Under the section of emotional attachment participants were presented with several different statements. The participants had to rank each statement on a scale from 1 (strongly disagree) to 7 (strongly agree) that best represented their beliefs and feelings about each question. Once they had completed the booklet a \$25 and a thank you note was sent.

Data Analysis

The data was analysed using an independent two-sample t-test under the assumption of normality and independence of observation between male and female group and within the groups. The variable used in this data analysis is one continuous variable and one binary variable. The continuous variable of interest to our independent sample t-test is love/attachment scale ranging from 1 indicating a lower score on love/attachment to 7 indicating a higher score on love/attachment with the binary variable of sex responses from males and females.

An independent sample t-test is appropriate for this data as it satisfies three main assumptions. The data included participants from a random digit dialing sample. The sample of females were selected independently from the sample of males. The two samples were in no means paired in a meaningful observation. To check for normality of each sample a Shapiro-Wilk test was performed on each sample. The female sample group and male sample group had a p-value > 0.05 indicating sample data comes from a normal distribution. The male sample group had a p-value close to 0.051. We would need to move further with a precaution

when performing the test and analysing the result. These test results of Shapiro-Wik can be observed in Figure A1 and Figure A2 for males and females respectively. For this test we assume that the samples have a homogeneity of variance.

Results

The variable of focus for our test hypothesis is the psychological characteristic of love/attachment. To understand any difference of love/attachment between the groups of females and males we test our hypothesis using independent two sample t-test. The hypothesis that we want to examine in this dataset is to observe any evidence for females scoring higher than males on the psychological characteristics of love/attachment. With this test we would be able to present evidence that females tend to have higher levels of love/attachment than males.

MIDUS 3 dataset consisted of 170 participants with 97 female and 73 male participants that completed the survey. The dataset contained a 57.1% response rate of females and 42.9% response rate of males. The mean value and standard deviation of the female group within the variable of love/attachment was reported 5.13 and 0.69 respectively. The mean value and standard deviation of the male group within the variable of love/attachment was reported 4.78 and 0.73 respectively. These results can be observed in Figure A3. The t-test reported a value of 3.15 for t statistics with a degree of freedom of 147. The p-value was reported <0.05 . With a small p-value there is strong evidence that the mean score of females on the psychological characteristics of love/attachment is higher than the mean score of males. The 95% confidence interval of the difference between females and males in the psychological characteristic of love/attachment was reported between 0.17 and infinity.

Discussion

Females on average score higher than males on the psychological characteristics of love/attachment. Our test provided further evidence to support our hypothesis. Although there was a statistical significance observed in our test between our two groups, female and male, when assessing the results of this test we must be cautious of the normality of our male group. Further when looking at the result we must be cautious that these groups are only a sample under our assumption that these samples are representative of our population of females and males to better understand how females and males score average on psychological characteristics of love/attachment. Using this finding we can better understand the psychology and the different behavior of females and males when exposed to different experiences. To have more supporting concrete evidence for our hypothesis we would need to replicate our test in different countries and among different age groups to get a better understanding of love/attachment difference. When conducting and replicating such a type of study one must keep in mind the way one participant can rate love/attachment differently than another person since the variable of love/attachment is not an objective score but rather subjective for each participant taking part in a study. With the help of this study we can aim to understand how women and men interact differently with each other to form relationships and how one maintains these relationships with the help of examining the fundamental characteristics of emotional attachment. Examining emotional attachment can help us understand romantic relationships, family dynamics and societal functioning.

References

- Abdel-Khalek, A. M., Lester, D., Dadfar, M., Atef Vahid, M. K., El Nayal, M. A., Alhuwailah, A., Zine El Abiddine, F., Singh, A. P., & Turan, Y. (2023). An examination of culture and gender differences on the Love of Life Scale (LLS) and its psychometric properties. *Mental Health, Religion & Culture*, 26(6), 591–599.
<https://doi.org/10.1080/13674676.2022.2034772>
- Fischer, A., & LaFrance, M. (2015). What Drives the Smile and the Tear: Why Women Are More Emotionally Expressive Than Men. *Emotion Review*, 7(1), 22–29.
<https://doi.org/10.1177/1754073914544406>
- Johnston, C. C., Campbell-Yeo, M., & Filion, F. (2011). Paternal vs Maternal Kangaroo Care for Procedural Pain in Preterm Neonates: A Randomized Crossover Trial. *Archives of Pediatrics & Adolescent Medicine*, 165(9), 792–796.
<https://doi.org/10.1001/archpediatrics.2011.130>
- Lein, V., University of Wisconsin Survey Center, National Institute on Aging, University of Wisconsin Institute on Aging, University of Wisconsin, Ryff, C., Schaeffer, N. C., Stevenson, J., Elver, K., Coombs, S., Sanchez, G., White, E., Salick, A., Day, B., Golan, K., Lawrence, D., Breen, R., Matoushek, J., Schultz, R., . . . Schultz, L. (2015). Midlife in the United States (MIDUS) Field Report.
https://midus-study.github.io/public-documentation/M3P1/Documentation/M3_P1_Field_Report_20220531.pdf
- Nelson, S. M., & Rubin, S. (1997). Sex differences, parental attachment, and juvenile delinquency. *Journal of Police and Criminal Psychology*, 12(2), 28–34.
<https://doi.org/10.1007/BF02806699>

Neto, F. (2023). Gender Differences in Self-Estimated Types of Love for Self and Others.

Interpersona: An International Journal on Personal Relationships, 17(1), 130-142.

<https://doi.org/10.5964/ijpr.9297>

Okubo, K. (2024). Parental Caregiving and Adolescent Attachment as Determinants of

Adolescent Self-Improvement: Differences Between Mother– and Father–Child Dyads in

Japan. The Family Journal (Alexandria, Va.), 32(2), 249–262.

<https://doi.org/10.1177/10664807231193438>

Understanding Midus samples. (n.d.).

https://midus.wisc.edu/findings/Understanding_MIDUS_Samples_20241024.pdf

Appendix

Figure A1

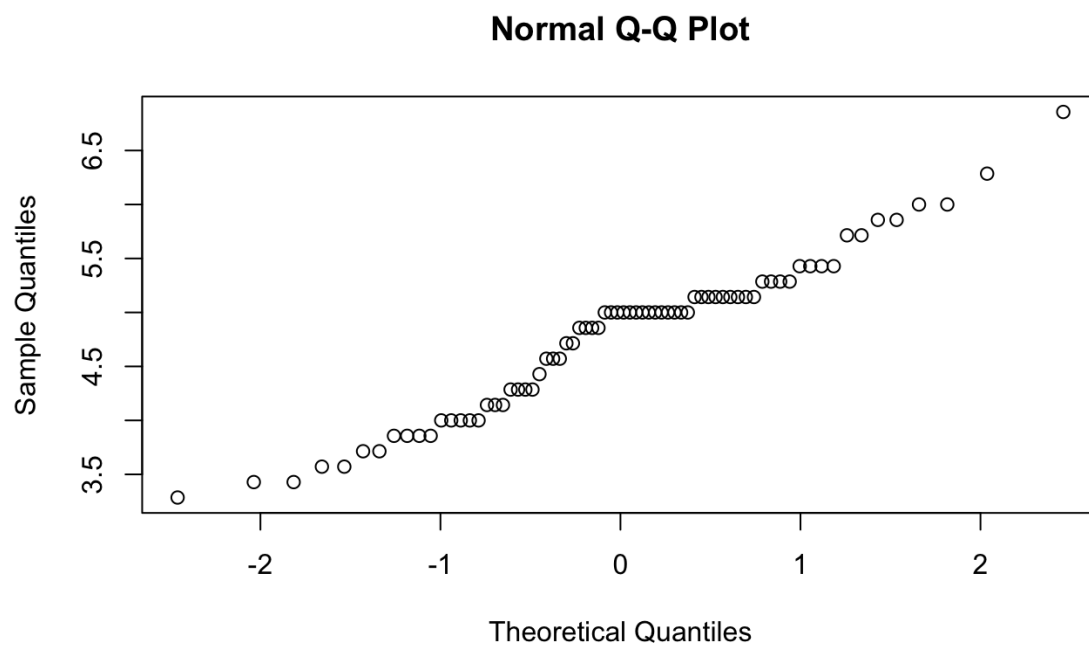
Normal Probability Plot for Males

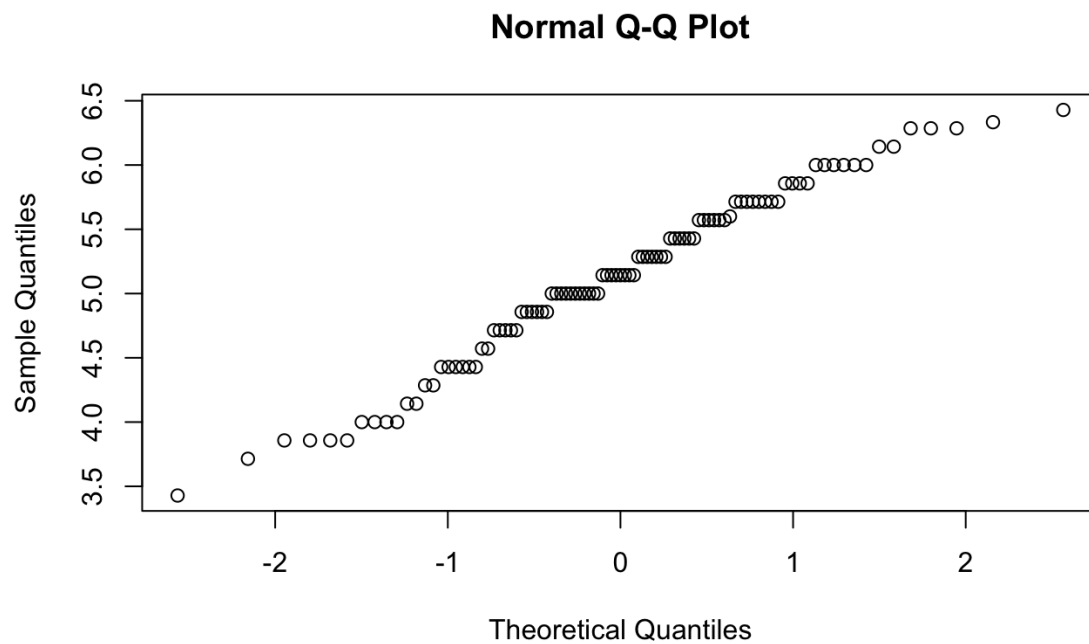
Figure A2*Normal Probability Plot for Females*

Figure A3*Comparative Density Histogram of Males and Females*